

Language generation from images

- A pure LLM is trained with causal language modeling for next token prediction:

- Maximize $P(w_t \mid w_1, w_2, \dots, w_{t-1})$

$$\mathcal{L}_{batch} = \frac{1}{B} \sum_{i=1}^B \sum_{t=1}^{T_i} -\log P_{\theta}(x_t^{(i)} \mid x_1^{(i)}, \dots, x_{t-1}^{(i)})$$

- VLM for text:
 - Align visual and textual representations
 - Generate text conditioned on images
 - Prefixing tokens in transformers
 - Instructional tuning through VQA
 - Adapter layer -> prefixing, deep mixing

How to generate text from images?

- Older methods focused on label generation beyond classification
 - Zero-shot learning
 - Aligning visual and textual representation
- Basic approach:
 - Image encoder
 - Language decoder
 - Transition from image to language (during training)
 - By prefixing image tokens to language tokens
 - Cross-attention where the language model attends to visual tokens looking back at relevant parts of the image
 - By mapping from image to language through an adapter layer
- Training datasets:
 - Image-label, Image-caption datasets
 - Instruction following multimodal datasets
 - VQA

Aligning visual textual representation

- Aligning visual-textual representations
 - Zero-shot learning
 - Contrastive loss (e.g. CLIP, SimCLR)
 - Image-text matching (e.g. in **BLIP**, **UNITER**, **ViLBERT**, **FLAVA**)
 - (image, text) → Match/NoMatch
- Multimodal language modeling:

$$P(\text{text} \mid \text{image})$$



input



A dog in a park

Target

- Masked language modeling with vision context: (**ViLBERT**, **LXMERT**, **UNITER**, **FLAVA**)



A - is sitting on a table



cat

How are pure LLM for text trained?

- Maximize $P(w_t \mid w_1, w_2, \dots, w_{t-1})$

Next token prediction through softmax and linear layer $z_t = W_{out} \cdot h_t + b$

- The model produces logits $z_t \in \mathbb{R}^V$ for a vocabulary of size V .
- Probabilities via softmax:

$$P_{\theta}(x_t = v \mid x_{<t}) = \frac{\exp(z_{t,v})}{\sum_{v'=1}^V \exp(z_{t,v'})}$$

- Loss uses the log of the probability for the **ground truth token** x_t .

$$\mathcal{L}_{batch} = \frac{1}{B} \sum_{i=1}^B \sum_{t=1}^{T_i} -\log P_{\theta}(x_t^{(i)} \mid x_1^{(i)}, \dots, x_{t-1}^{(i)})$$

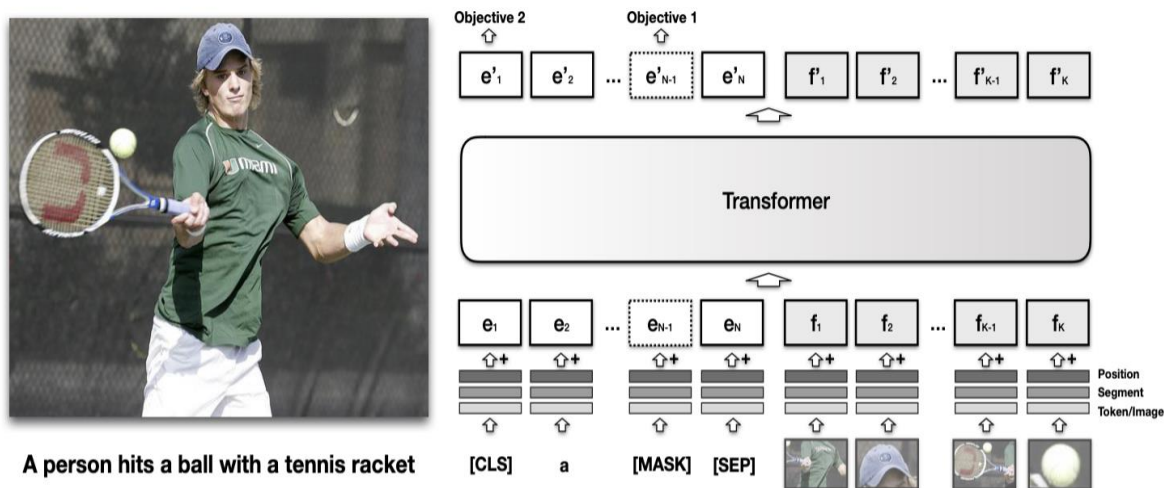
Minimizing this loss is equivalent to **maximizing the likelihood of the observed data**:

$$\hat{\theta} = \arg \max_{\theta} \prod_{i=1}^B \prod_{t=1}^{T_i} P_{\theta}(x_t^{(i)} \mid x_{<t}^{(i)})$$

MLE

VLM- Combining image and text in a transformer

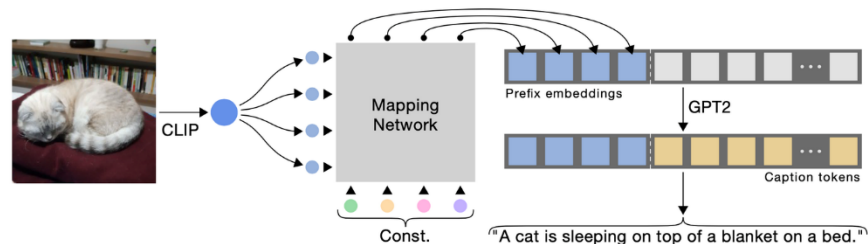
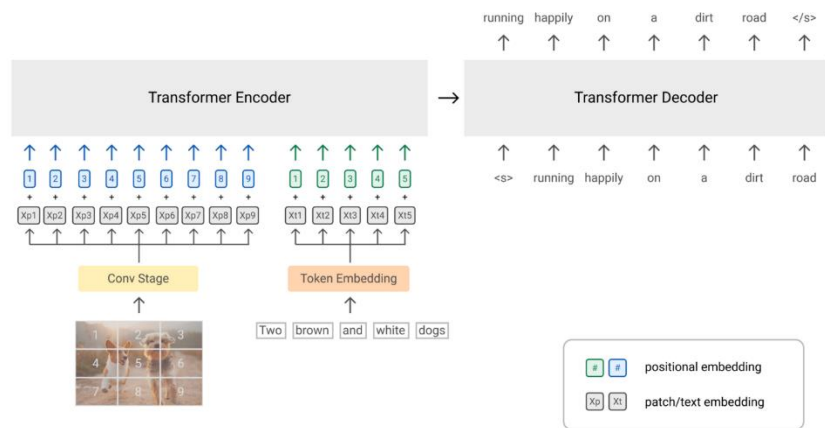
Translating images into embedding features that can be jointly trained with token embeddings.



L. H. Li et al. VisualBERT: A Simple and Performant Baseline for Vision and Language, ACL 2020

VLM- Vision prefix for an LLM

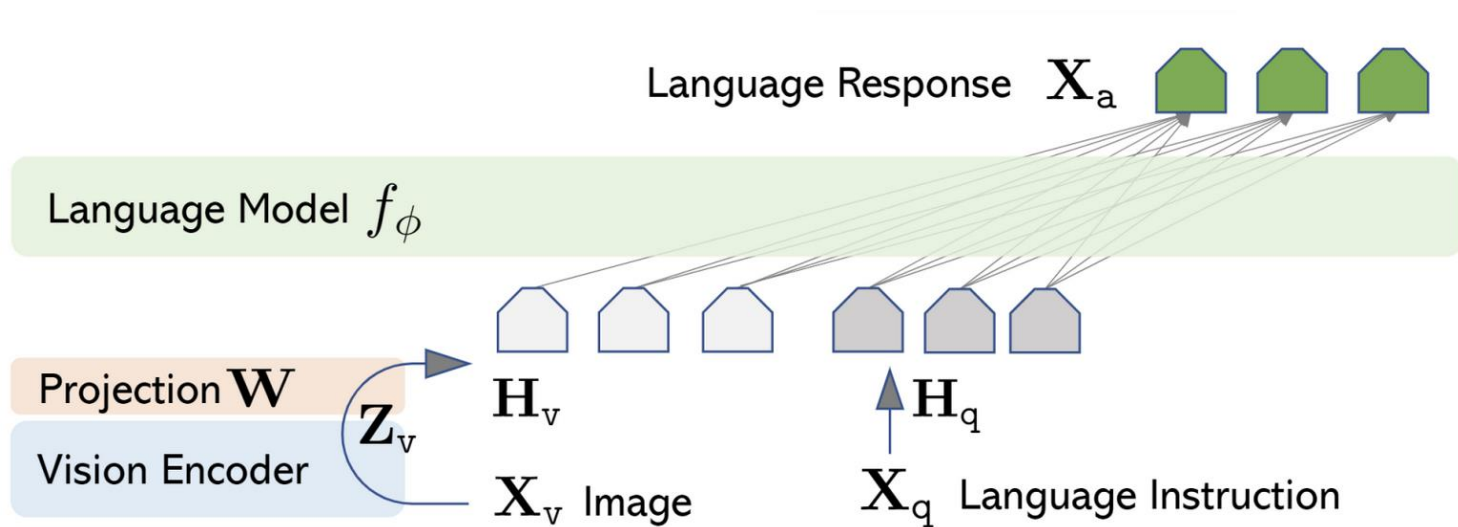
Learning good image embeddings that can work as a prefix for a frozen, pre-trained language model.



Frozen, CLIPCAP, language models with vision encoders

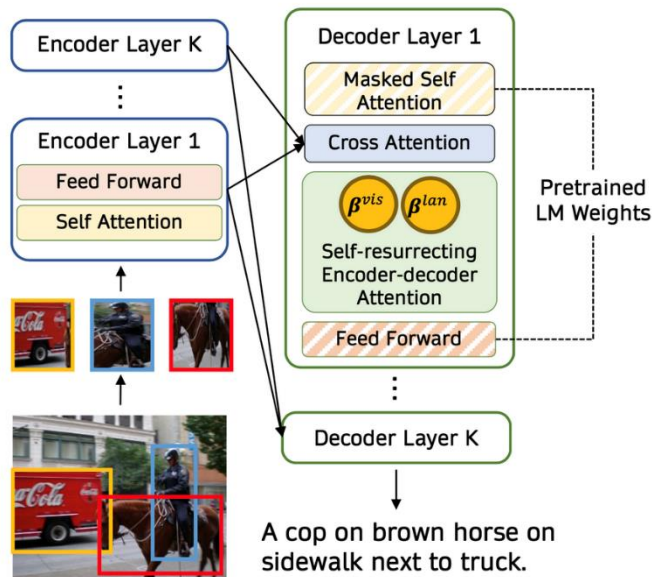
2019 -2021

Llava Model

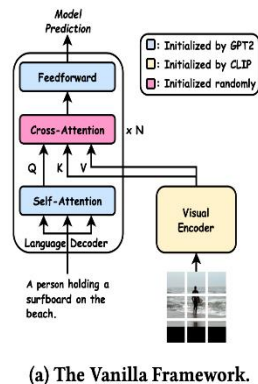


Visual Instruction Tuning, NeurIPS2023

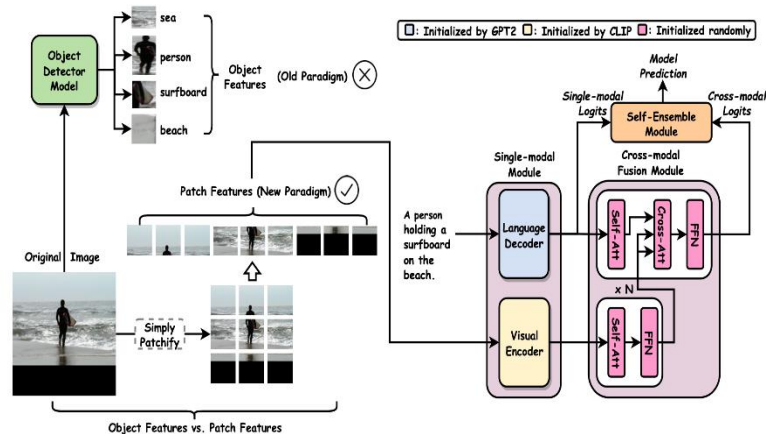
VLM - Text-Image Cross-Attention Fuse Mechanisms



VisualGPT



(a) The Vanilla Framework.

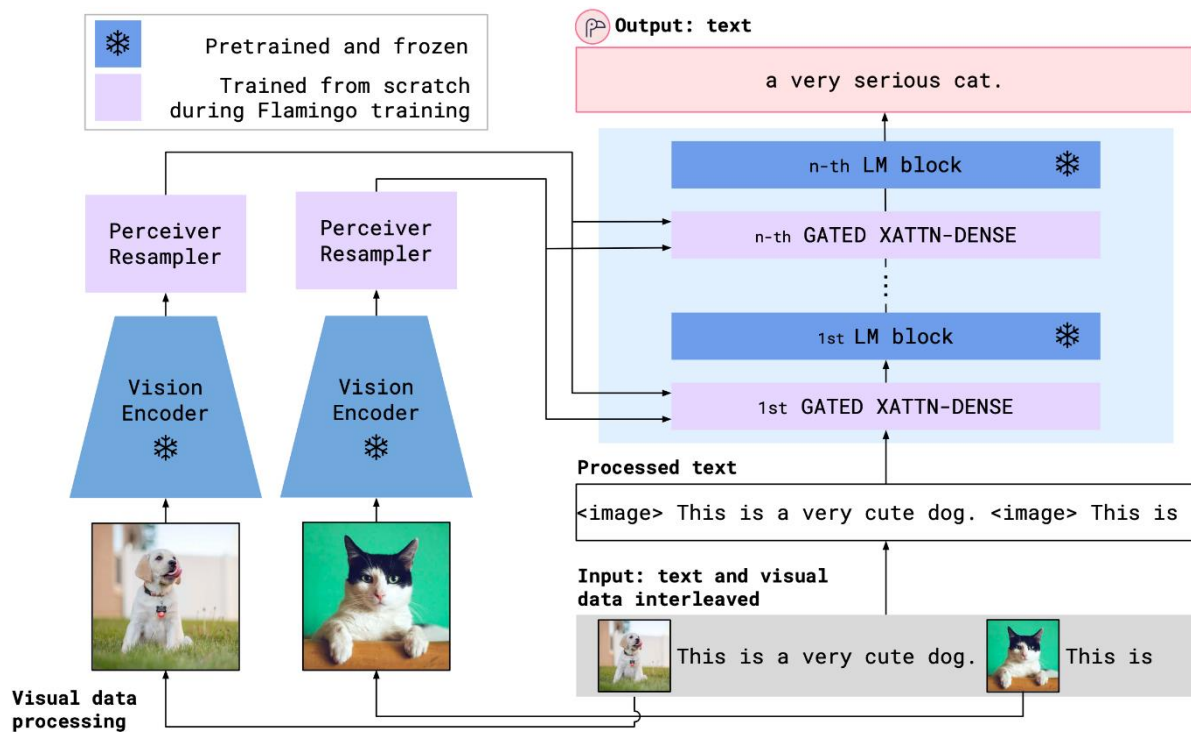


(b) Our End-to-End VC-GPT Framework.

VC-GPT

VLM - Multimodal Generative AI

Flamingo



Interleaved placement of multimodal input

What does the adapter do?

- **Dimensional mismatch** between visual embeddings and LLM token embeddings.
- **Information selection**, by letting the LLM attend only to the most relevant visual features.
- **Low-data learning**, allowing instruction-following behaviors to emerge without massive retraining.

Generating text conditioned on images

- Prefixing tokens in transformers
- Instructional tuning through VQA
- Adapter layer

(Image + Instruction text) → Response text

User: What is the person holding?

<Image>

Assistant: The person is holding a tennis racket.

Typical modern VLM training

- Gemini, Llama, Granite, etc. follow Llava paradigm of projecting image to text through an MLP layer
 - Freeze LLM, vision encoder but train MLP
- Do vision language alignment using image-text captions
 - E.g. LIAON-5B captions for alignment
- Continue with multimodal instruction tuning

User: What **is** the person holding?

<Image>

Assistant: The person **is** holding a tennis racket.

Prefixing (Llava)

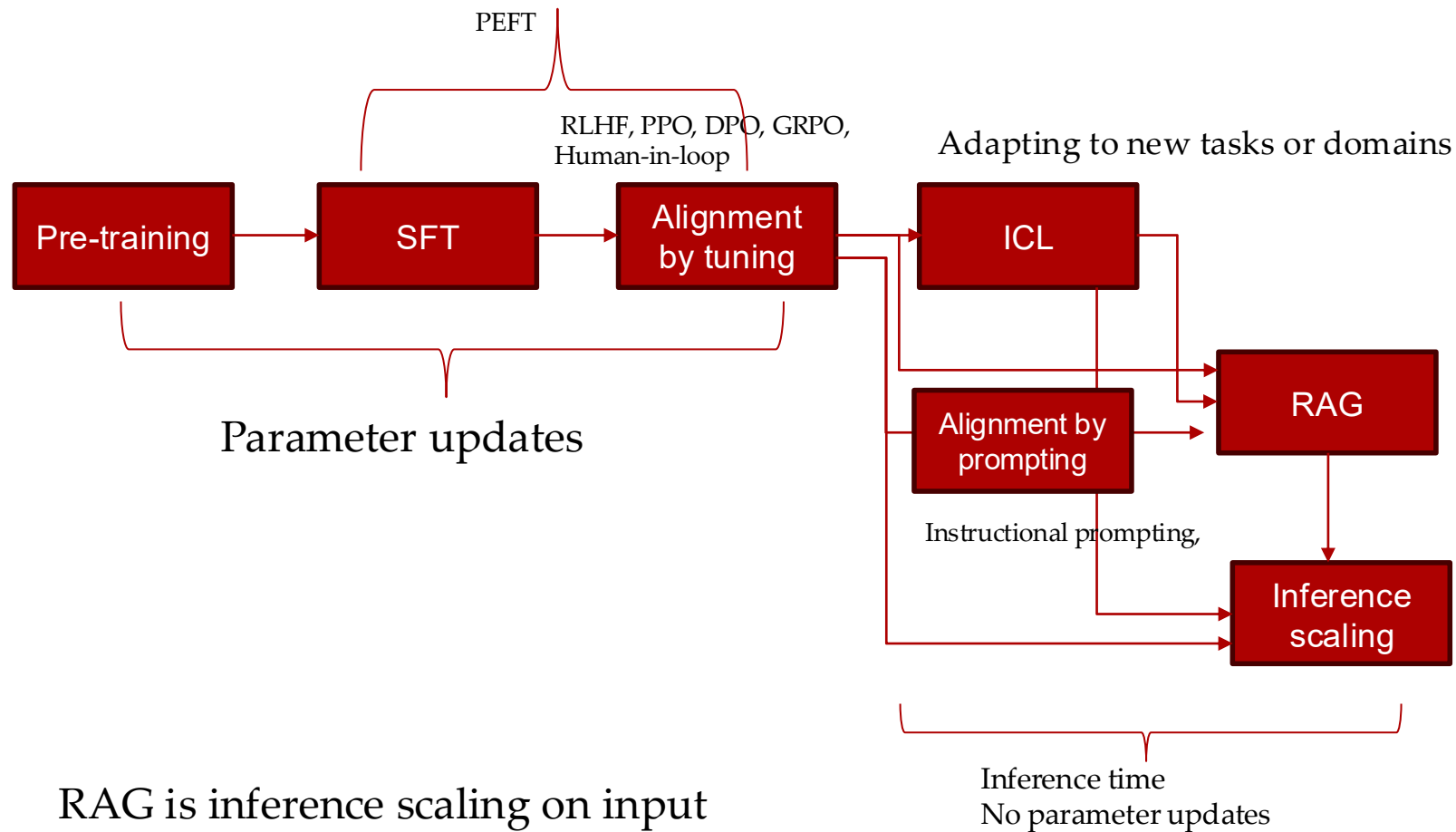
[<img_tokens>] [<instruction_text_tokens>]

Cross-attention
(Flamingo)

Steps in training LLMs/VLMs

- 3 steps of training LLMs
 - Self-supervised training
 - Training a base model on a vast amount of unlabeled text data using self-supervised objective, such as predicting the next token
 - SFT is **not** primarily about raw knowledge or capability discovery—that mostly comes from pretraining.
 - Supervised fine-tuning
 - Training on smaller, labeled dataset for specific downstream tasks (e.g., sentiment analysis, summarization, or instruction following). Fine-tuning modifies the model's parameters to improve performance on these specialized tasks.
 - Alignment for improving performance
 - Post-training and fine-tuning phase for shaping the model's behavior, safety, and ethical compliance
 - With or without parameter tuning
 - Based on human preferences
- **Supervised Fine-Tuning (SFT)** is used to teach the model:
 - How to *behave* as an assistant
 - How to map **instructions** → **desired responses**
 - Formatting, tone, structure, refusals
 - Canonical ways to solve problems *when examples exist*

LLM training and improvement



Concept subtlety

- Alignment vs. scaling
 - Alignment: can modify parameters, goal is to **shaping the model's behavior to match human-desired values, instructions, or safety constraints**.
 - Scaling: improves accuracy, improves reasoning
- Prompting during alignment vs during scaling vs ICL
 - Prompting during alignment: Guide the model to **behave in line with human preferences, values, or safety rules**.
 - Prompting during ICL: Adapt the model to a **new task** at **inference time** without updating parameter
 - Prompting during scaling: Enhance model capability temporarily by **exploiting latent capacity**.
- How come no parameter updates still improve performance?

How does improvement happen if no parameters are updated?

- Prompting changes the output of LLM acting as a steering mechanism that shifts the probability distribution of predicted tokens, guiding the model toward specific, relevant, and accurate responses.
- After seeing examples like this, hidden states tend to land in regions that support the correct continuation.

Alignment methods summary

Method	Parameter updates?	Mechanism	Strengths	Weaknesses
Instruction-tuning / SFT	✓	Supervised fine-tuning on desired behavior	Direct task shaping	Requires labeled data
RLHF	✓	Reward model + RL	Strong alignment	Computationally heavy
DPO	✓	Preference-based optimization	Cheaper than RLHF	Less flexible than full RLHF
Rule-based / filters	✗	Post-processing, hard constraints	Safety guarantees	Circumventable, doesn't change reasoning
Instruction prompts	✗	Prompting / ICL	No retraining needed	Soft, brittle
Constitutional AI / self-critique	✗/✓	Model revises own output	Scales oversight	Depends on base capabilities
Human-in-the-loop	✓	Feedback integration	High-quality alignment	Expensive, manual

Alignment versus inference scaling

- **Alignment:**
 - Modify the model's parameters
 - Change the probability distribution *globally*
 - Affect behavior across all prompts
- **Inference scaling** keeps the model frozen and instead:
 - Uses more tokens
 - Uses more samples
 - Uses search, voting, reflection, or verification
 - Inference targets:
 - Accuracy
 - Reasoning depth
 - Robustness to randomness
 - Fewer arithmetic or logic mistakes

Alignment versus Scaling

Dimension	Alignment	Inference Scaling
Changes weights?	✓ Yes	✗ No
Happens when?	Training time	Inference time
Affects future queries?	✓ Yes (global)	✗ No (ephemeral)
Main goal	Behavior & safety	Accuracy & reasoning
Cost	High upfront	High per query
Reversible	✗ No (unless retrained)	✓ Yes
Typical failure	Over/under-refusal, bias	Hallucinated reasoning
Scope	Global policy	Single response

Instructional prompting vs. scaling

Important contrast:

- **Instruction prompting** biases *what the model tries to do*
- **Inference scaling** biases *how hard it tries*

Instruction prompting:

“Be safe, be helpful, follow rules.”

Inference scaling:

“Think longer before answering.”

They operate on different axes.

Instructional prompting *is* a form of alignment, but it is **conditional and fragile**, whereas parameter-based alignment is **global and robust**; modern systems rely on both.

COT : Is this an alignment or scaling method?

CoT does this:

- Encourages **explicit intermediate reasoning**
- Increases number of forward-pass steps
- Allows state refinement before committing to an answer
- Improves correctness on multi-step tasks

It answers:

How should the model compute the answer?

That is **inference scaling**, not alignment.

A misaligned model with CoT:

- Will still reason its way to a harmful answer
- Just more carefully and convincingly

Guess which method?

Method	Changes parameters?	Primary effect	Category
RLHF / instruction tuning	✓	Values & preferences	Alignment
System / instruction prompts	✗	Behavioral conditioning	Prompt-based alignment
Chain-of-thought	✗	More computation & reasoning	Inference scaling
Self-consistency	✗	Error reduction	Inference scaling
Tree-of-thought	✗	Search	Inference scaling
Refusal templates	✗	Behavioral constraints	Prompt-based alignment
Output filtering	✗	Hard safety bounds	Alignment (external)

Improving LLM performance post-training and fine-tuning

- **Alignment changes *what the model wants to do*** (its preferences and constraints),
 - **Alignment** is about making the model's outputs:
 - Helpful
 - Harmless
 - Honest
 - On-policy (follows instructions, refuses when needed)
- **Inference scaling changes *how hard the model thinks*** before answering—without changing its preferences or knowledge.
- Alignment methods
 - Instruction tuning
 - RLHF / RLAI
 - Preference optimization
 - Constitutional AI-style methods
- Without parameter tuning
 - Prompt engineering
- With parameter tuning
 - Reinforcement learning with human feedback
- RAG
- MoE

Prompt Engineering

- Zero-shot prompting
- Prompt chaining
- Chain of thought
- Object-guided prompting
- Meta prompting

Prompts do matter:

- E.g. Different results with slightly different prompt
 - A photo of a plane
 - A photo of a plane parked at the gate
 - A photo of a plane parked at the gate or flying in the sky

The art of prompting

Be Specific: Prompt should be specific

Indicate Intent: Make sure your intent is clear

Show Examples: use them for formatting and inferring commonalities

Given Clear Instructions: The instructions shouldn't be ambiguous

Indicate Desired Output: Indicate the format of the output

Elements of a prompt

A prompt contains any of the following elements:

Instruction - a specific task or instruction you want the model to perform

Context - external information or additional context that can steer the model to better responses

Input Data - the input or question that we are interested to find a response for

- **Output Indicator** - the type or format of the output.

Few-shot prompting

- Is a type of "in-context learning" or "learning by example."
 - Pattern recognition:
 - identifying patterns in how inputs are transformed into outputs.
 - Task inference:
 - The nature of the task being asked to perform
 - Generalization:
 - Generalize from given examples to new inputs
 - Application:
 - Apply the learned pattern to new input
- What are some pitfalls?
 - What if there isn't enough variety?
 - What if there is too much variety?

Few shot example:

- Classify the sentiment of the third movie review. Use the information from the first two examples:
- Review: "This movie was a waste of time.
- "Sentiment: NegativeReview:
- "I couldn't stop laughing throughout the film!
- "Sentiment: PositiveReview:
- "The special effects were amazing, but the plot was confusing
- "Sentiment:"

Provide a possible diagnosis and explain your reasoning:

Symptoms: Fever, cough, fatigue

Diagnosis: Common cold

Explanation: The combination of fever, cough, and fatigue is typical of a common cold. No severe symptoms are present, suggesting a mild viral infection.

Symptoms: Chest pain, shortness of breath, dizziness

Diagnosis: Possible heart attack

Explanation: The combination of chest pain, shortness of breath, and dizziness are warning signs of a possible heart attack. Immediate medical attention is required.

Symptoms: Headache, sensitivity to light, nausea

Diagnosis:

Explanation:

In-context learning versus training

In-context learning can implement behaviors that **gradient updates can't**, such as:

- Conditional policies (“for *this* task only”)
- Context-dependent definitions
- Temporary role-play or personas
- One-off transformations

• Training changes the model everywhere.

Examples where incontext training is used in combination :

- **Adapters / LoRA hot-swapping**
- **Test-time training** in vision models
- **Reinforcement learning from interaction**
- **Memory-augmented models**
- **User-specific personalization layers**

These are:

- Carefully sandboxed
- Often session-scoped
- Usually slow
- Rarely exposed directly to raw user prompts

So the idea isn't rejected—it's just tightly

In-context learning tradeoffs

Property	In-Context Learning
Speed	✓ very fast
Safety	✓ stateless
Flexibility	✓ extremely high
Cost	✓ low
Guarantees	✗ none (but acceptable)
Stability	⚠ probabilistic

In context learning

- Why not do incontext training?
 - Splitting the prompt to input and output is not easy
 - Catastrophic overfitting could happen
 - Backprop is way more costly than forward pass (speed and cost)
 - Safety and predictability
 - Prompt injection , model drift, data contamination

Chain-of-thought prompting

- Tackle address complex problems by breaking them into simpler, intermediate steps and encouraging the model to “think” before answering

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

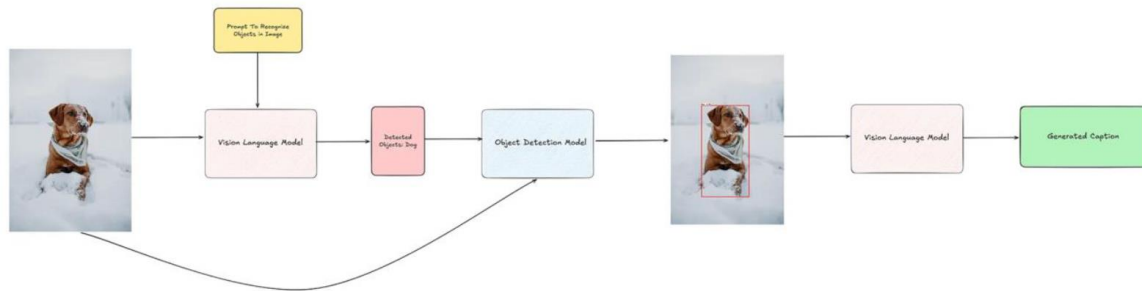
Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Object-guided prompting

- Combined open vocabulary tagging with object detection to generate a more focused caption

system_prompt=""You are a helpful assistant that can analyze images and provide captions. You are provided with images that also contain bounding box annotations of the important objects in them, along with their labels.

Analyze the overall image and the provided bounding box information and provide an appropriate caption for the image.""",



<https://medium.com/data-science/prompting-with-vision-language-models-bdabe00452b7>

Meta-prompting

- Focuses on the structure rather than infer from the answer unlike COT

Integrate step-by-step reasoning to solve mathematical problems under the following structure:

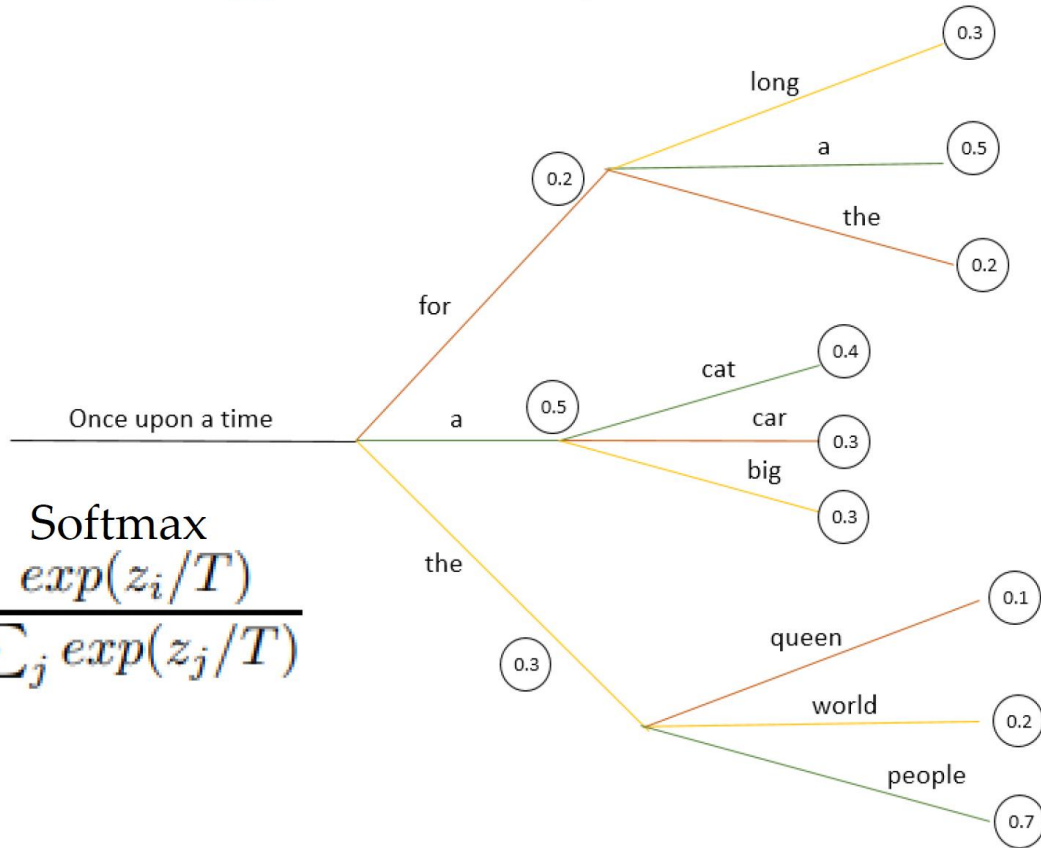
```
{
  "Problem": "[question to be answered]",
  "Solution": {
    "Step 1": "Begin the response with 'Let's think step by step.'",
    "Step 2": "Follow with the reasoning steps, ensuring the solution process is broken down clearly and logically.",
    "Step 3": "End the solution with the final answer encapsulated in a LaTeX-formatted box,  $\boxed{\dots}$  for clarity and emphasis."
  },
  "Final Answer": "[final answer to the problem]"
}
```

[Meta Prompting for AI Systems](#)

Understanding the LLM text generation process

- Transformer decoder
- Next token selector
 - Beam search
 - Deterministic
 - Stochastic
 - Random
 - Temperature-based
 - Top-k
 - Top-p

$$q_i = \frac{\text{Softmax} \exp(z_i/T)}{\sum_j \exp(z_j/T)}$$



Exercise: Guess prompting type

Classify the text into neutral, negative or positive.

Text: I think the vacation is okay.

Sentiment:

A "whatpu" is a small, furry animal native to Tanzania. An example of a sentence that uses the word whatpu is:

We were traveling in Africa and we saw these very cute whatpus.

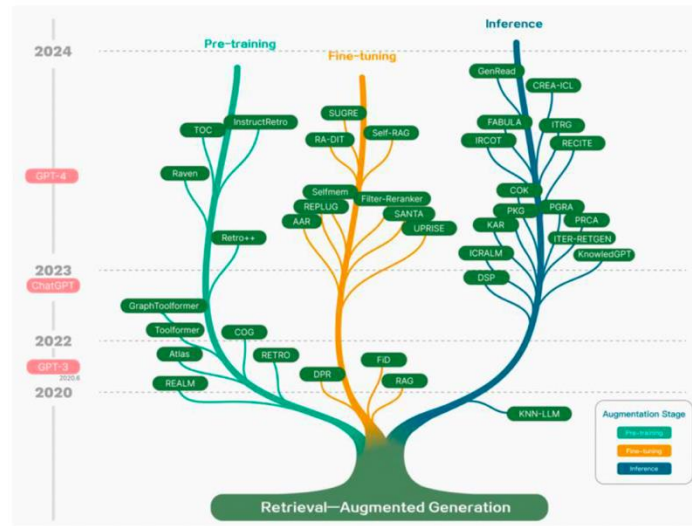
To do a "farduddle" means to jump up and down really fast. An example of a sentence that uses the word farduddle is:

Here are the rules for question and answer generation. 1) The question should not be a multiple choice question and answer. 2) The answers should be in a single paragraph (no bullet points). 3) The questions should be tagged as Question: and the answers should be tagged as Answer: 4) Do not generate any other text before and after the questions and answers. 5) If you are unable to generate question and answers your response should be - Unable to generate questions and answers. 6) Do not repeat the same question. Using these rules, generate 5 questions and answers based on the following text:

<https://arxiv.org/abs/2005.14165>

Retrieval augmented generation

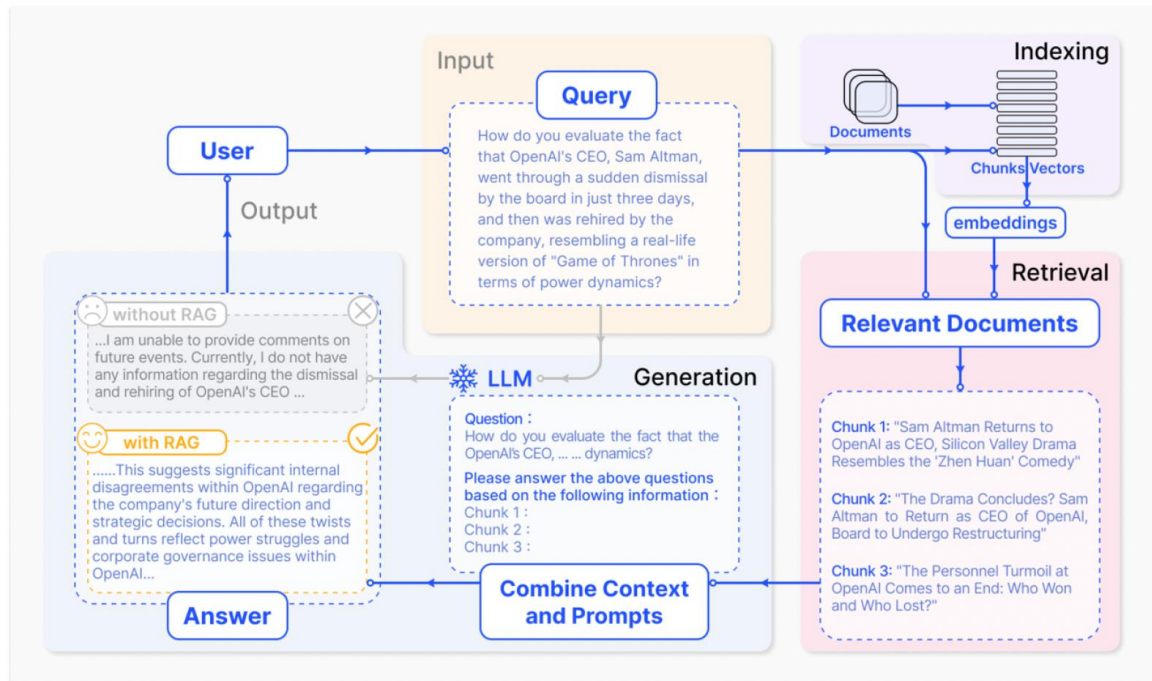
- A way to improve the performance of LLM
 - Could be used during pre-training or fine-tuning but mostly used during inference
- Practical approach to improve the model response with latest onsite knowledge
- Minimizes hallucinations with provided knowledge
- Time-relevant responses
- Increases transparency and gives traceable reasoning
- Cost-effective way to deploy models
- Pivotal technology for chatbots and practical applications.
- Widely implemented in commercial applications and popularized vector databases



Retrieval-Augmented Generation for Large Language Models: A Survey, Jan 2024

RAG – Key idea

- Given a query to an LLM
 - Search/retrieve relevant documents
 - Construct prompts with the relevant documents
 - LLM to generate revised response
- How to determine relevancy?
- How does the new information integrate into LLM?
- How to evaluate the improvement?



RAG – the formulation

Given a query x , the retriever selects the top- k documents $\{d_1, \dots, d_k\} \subset \mathcal{D}$ using a similarity function, defined as

$$\text{sim}(x, d_i) = f_q(x)^\top f_d(d_i) \quad (1)$$

where f_q and f_d are neural encoders for the query and documents, respectively. Similarity is computed using the dot product or cosine similarity in a shared embedding space.

The generator produces output y by estimating the conditional probability over possible responses:

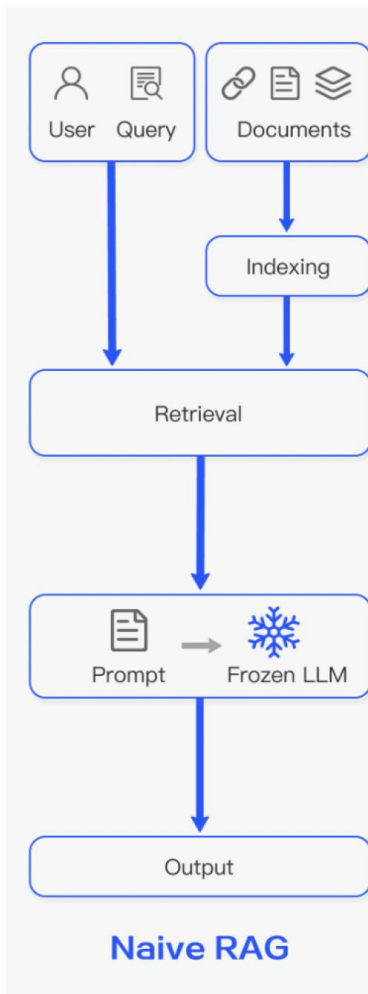
$$P(y \mid x) = \sum_{i=1}^k P(d_i \mid x) \cdot P(y \mid x, d_i) \quad (2)$$

Here, $P(d_i \mid x)$ can be derived from normalized similarity scores or treated uniformly in top- k retrieval. The final output can be generated either by marginalizing across retrieved documents or by concatenating them as a single input:

$$y = \arg \max_{y'} P(y' \mid x, \{d_1, \dots, d_k\}) \quad (3)$$

Naïve RAG

- Indexing
 - Text conversion
 - Chunking documents
 - Encoding documents
 - Index generation (k-NN), e.g. Faiss index
- Retrieval
 - Cosine similarity retrieval by encoding query
- Generation
 - Responses to the information contained within the provided documents
 - In cases of ongoing dialogues, any existing conversational history can be integrated into the prompt,
 - enabling the model to engage in multi-turn dialog



- Problems:

- Low precision and recall
- Irrelevant context in responses
- Augmentation from the retrieved chunks tricky due to redundancies and repetition
- Novel content generation may not be possible

Search/Retrieval methods for RAG

- Unsupervised approaches
 - Sparse retrieval
 - Dense retrieval
 - Sparse vector retrieval
- Supervised approaches
 - Neural IR
 - Re-ranking models

Lexical or Sparse Retrieval Methods - Unsupervised

- Based on traditional document retrieval approaches
- Breaks up a document and query into tokens
- Match is determined using TF/IDF rules
 - Term Frequency (TF)
 - How often a term occurs in a document
 - Inverse Document Frequency (IDF)
 - How common is such occurrence across documents
- BM25 is a popular sparse retrieval method
 - Uses TF/IDF
 - Document Length Normalization
 - Measure should be robust to document lengths

Given a query Q , containing keywords $q_1, \dots, q_n$, the BM25 score of a document D is:

$$\text{score}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)}$$

where $f(q_i, D)$ is q_i 's **term frequency** in the document D , $|D|$ is the length of the document D in words, and avgdl is the average document length in the text collection from which documents are drawn. k_1 and b are free parameters, usually chosen, in absence of an advanced optimization, as $k_1 \in [1.2, 2.0]$ and $b = 0.75$.^[1] $\text{IDF}(q_i)$ is the IDF (**inverse document frequency**) weight of the query term q_i . It is usually computed as:

$$\text{IDF}(q_i) = \ln \left(\frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} + 1 \right)$$

where N is the total number of documents in the collection, and $n(q_i)$ is the number of documents containing q_i .

Sparse retrieval methods:

- Allow exact matching – both a boon and a curse
- Efficient inverted indexing

Dense or Semantic retrieval - Unsupervised

- Given a set of documents
 - Extract chunks from documents
 - Vectorize the chunks using an embedding
- Given a query:
 - Vectorize the query using the same embedding
- Find the closest match to the query using distance or similarity metrics:
 - Inner product
 - Cosine similarity
 - L2 distance
 - Softmax over relevant scores to convert to probability
- When the documents grow large, the vectors could grow to very large vector stores
 - Compression
 - Approximate nearest neighbor search. Through indexing

$$a \cdot b = \sum_{i=1}^n a_i b_i$$

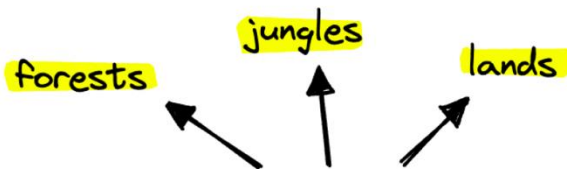
$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

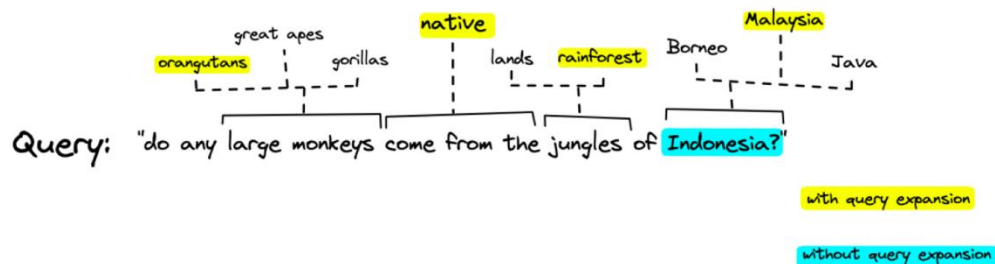
$$p(z | x) = \frac{\exp f(x, z)}{\sum_{z'} \exp f(x, z')},$$

$$f(x, z) = \text{Embed}_{\text{input}}(x)^\top \text{Embed}_{\text{doc}}(z)$$

Sparse learned embeddings for retrieval



"orangutans are native to the rainforests of Indonesia and Malaysia"



Doc: "Orangutans are native to the rainforests of Indonesia and Malaysia"

<https://www.pinecone.io/learn/splade/>

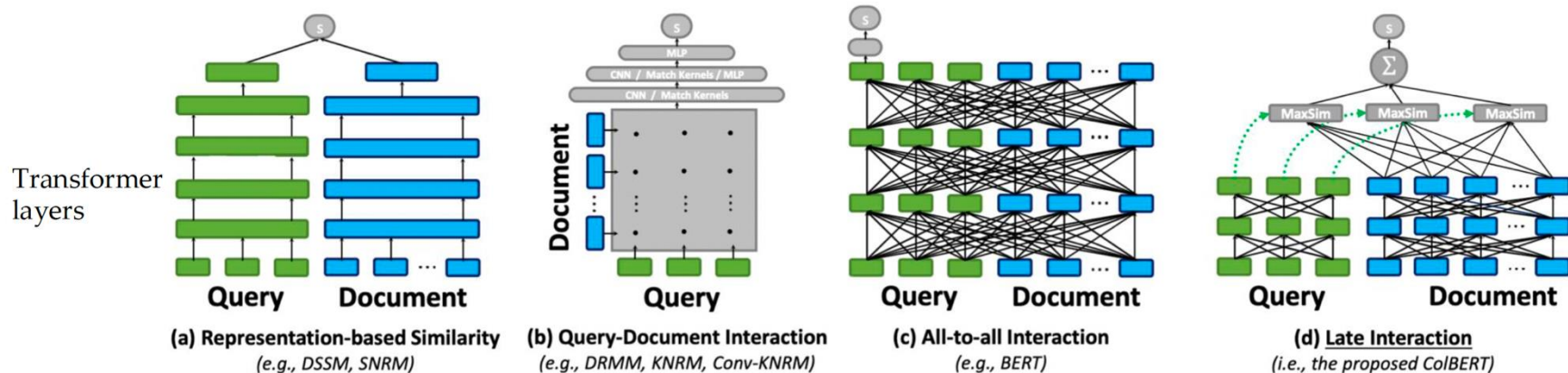
Document	SPLADE BOW rep	New terms (expansion)	Dropped terms (compression)
Binary (or base-2) a numeric system that only uses two digits—0 and 1. Computers operate in binary, meaning they store data and perform calculations using only zeros and ones.	(binary, 2.96), (base, 1.83), (computer, 1.57), (2, 1.42), (decimal, 1.29), (computers, 1.29), (system, 1.25), (digit, 1.24), (nu, 1.17), (0, 0.76), (zero, 0.74), (, 0.56), (the, 0.53), (two, 0.49), (harmony, 0.48), (operate, 0.4), (computing, 0.31), (digits, 0.24), (neon, 0.19), (pi, 0.17), (numeric, 0.14), (pc, 0.13), (, 0.06)	harmony, computing, digit, , the, pc, decimal, computer, pi, neon	—, perform, ##s, or, meaning, data,), (, store, in, 1, , only, calculations, and, , that, they, using, a, uses, ones
Option Types: Calls & Puts In the special language of options, contracts fall into two categories - Calls and Puts. A Call represents the right of the holder to buy stock.	(option, 2.16), (call, 1.92), (type, 1.75), (contract, 1.58), (put, 1.33), (language, 1.24), (holder, 1.18), (right, 0.95), (putting, 0.89), (deal, 0.83), (puts, 0.71), (options, 0.66), (special, 0.61), (buy, 0.6), (stock, 0.53), (contracts, 0.36), (calls, 0.34), (stuff, 0.25), (called, 0.14), (two, 0.05), (specialized, 0.04), (types, 0.02)	type, contract, put, deal, called, specialized, putting, stuff	into, in, of, , the, , and, , a, categories, &, fall, to, , represents

<https://europe.naverlabs.com/blog/splade-a-sparse-bert-encoder-bert-based-model-achieves-effective-and-efficient-first-stage-ranking/>

SPLADE v2: Sparse Lexical and Expansion Model for Information Retrieval, SIGIR 2021

Neural IR approaches

The supervised approach uses IR data such as labeled query-document pairs, to learn a representation that is optimized for ranking and retrieval



Query and document embeddings separately computed and compared at search time using cosine similarity

An interaction matrix that reflects the similarity between every pair of words across q and d fed to a classifier for relevance. The similarity matrix is fed to classifiers

Use a transformer to match and decode to a similarity value

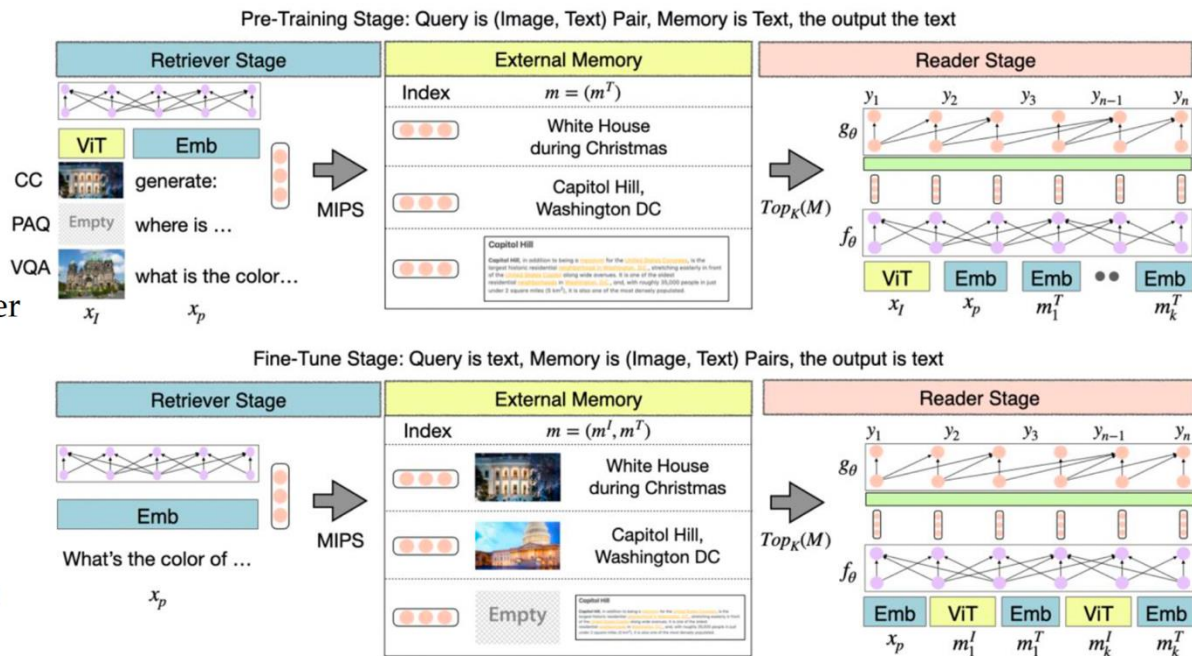
COLBERT

[Contextualized Late Interaction over BERT, SIGIR 2020](#)

MURAG – Multimodal RAG

<https://arxiv.org/abs/2210.02928>

- During pre-training: Train an encoder to produce relevant captions from a textual memory bank.
- During RAG: Use the same encoder for encoding image-text pairs in a multimodal memory bank.
- Input is a textual prompt, the retrieved documents are multimodal, which are then fed to the generator to generate text



Factors affecting RAG performance

- Extraction accuracy of the ingestion pipeline (structural shredders, chunkers, summarizers)
- How it got represented (a function of embedding choices and semantic expansions)
 - What representation capture semantics of documents?
 - What embedding spaces bring queries and documents closer?
- How it was searched (lexical, semantic, fusion, re-ranking)
- How it was aggregated and rolled up for the specific use case (chunk, page, document level roll-up)
- How the query was analyzed (semantic enrichment of the query)
 - Query rewriting
- Prompt engineering to prime the LLM
- Whether RAG is used only during inference or at pre-training or fine-tuning as well

Using RAG for healthcare applications

- *Diagnostic Assistance*
 - Similar patient cases
 - Retrieve relevant clinical guidelines
- Summarization
 - EMR, discharge notes
- Medical QA
- Patient education chatbots
- Clinical trial matching
 - *Aligning patient profiles with the eligibility criteria extracted from trial registries*
 - *Retriever identifies protocols, generator generates inclusion and exclusion criteria.*
- Biomedical literature synthesis
 - For research purposes

RAG applications in healthcare

Table 1. Taxonomy of RAG applications in healthcare.

No.	Category	Description
1	General Clinical Applications of RAG	Broad applications of RAG for tasks like clinical summarization, decision support, and guidelines.
2	RAG Chatbots for Patient Interaction	Conversational agents enhanced by retrieval for providing personalized medical advice.
3	Specialty-Focused RAG Models	RAG frameworks tailored for domains such as cardiology, nephrology, or oncology using specialty-specific knowledge bases.
4	RAG for Signal and Time-Series Tasks	Integration of RAG with biosignals like ECG, EEG, or wearable data for diagnostic interpretation.
5	Graph-Based and Ontology-Aware RAG Frameworks	Use of structured clinical ontologies or knowledge graphs for enhanced retrieval and explainability.
6	RAG with Blockchain and Secure Architectures	Incorporation of privacy-preserving, decentralized data retrieval using blockchain-enhanced architectures.
7	Radiology-Specific Retrieval-Augmented QA	RAG systems designed for image-report alignment, report generation, and visual question answering in radiology.

medical reasoning, discharge planning, and infectious disease support

patient self-management, pre-consult triage, mental health screening, telemedicine

specialty decision support, patient-specific education, and knowledge reinforcement

using time-series data enhance diagnostic accuracy by integrating structured physiological signals into prompts

keyword, vector, and graph retrieval with a validated knowledge graph using topic-based chunking and semantic matching

Secure and auditable retrieval in sensitive domains, e.g. pediatrics

Second opinions, structured reporting, and continuous medical education in complex cases

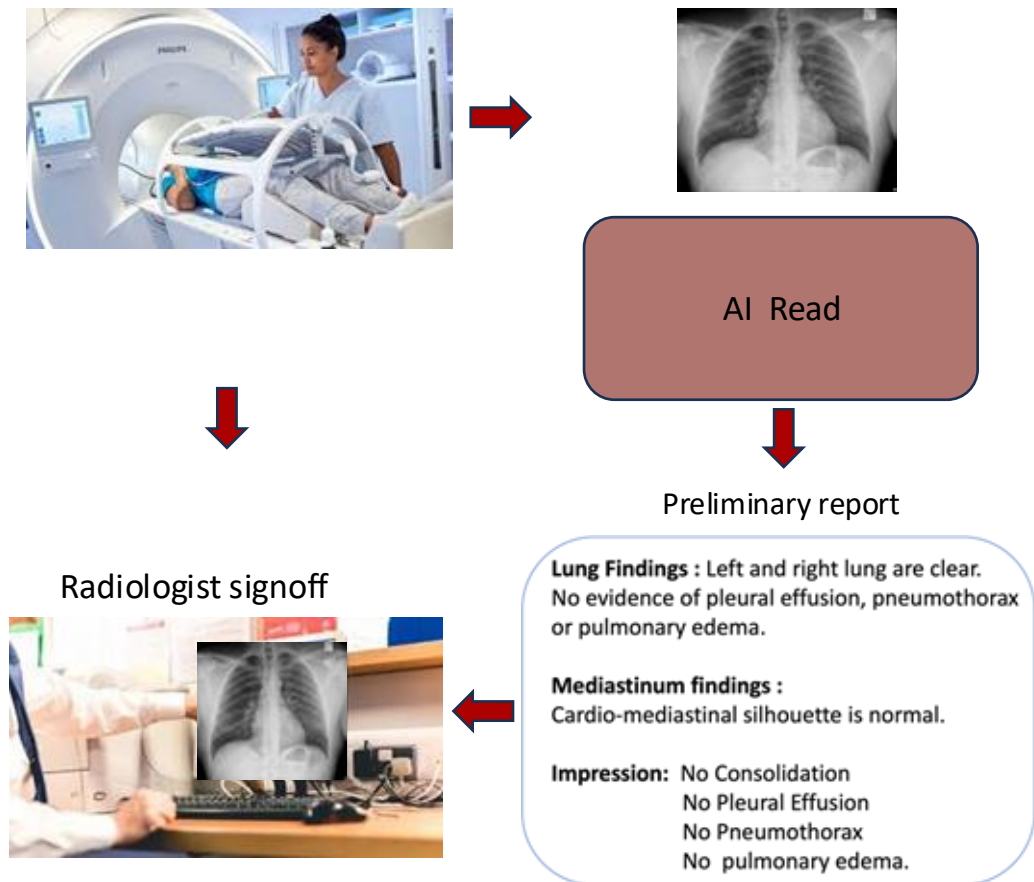
RAG Benchmarks in healthcare

- *MedQA (USMLE):*
 - *multiple-choice question dataset derived from medical licensing exams, focused on clinical knowledge assessment*
- *PubMedQA:*
 - *biomedical abstracts paired with yes/no/maybe questions, requiring grounded reasoning and evidence-based answers*
- *MIMIC-IV:*
 - *de-identified EHR dataset supporting tasks such as summarization, question answering, and document retrieval*
- *MedDialog:*
 - *multilingual dataset of doctor–patient conversations, suitable for training and evaluating medical dialogue systems*

RAG challenges in healthcare

- Effective use of clinical knowledge graphs in RAG
- Static versus dynamic RAG
 - updating retrieval indexes for evolving knowledge
- Multimodal integration still not prevalent
- Handling privacy issues
- Domain-specific evaluation metrics
- Clinical workflow integration
- Bias and fairness issues

Automated Radiology Report Generation (RRG) using Generative AI



- **Expedites clinical workflow operations**
 - Helps when experts are not available
 - Reduces radiologists' workload
 - Saves radiologists from dictation
 - Expedited review
 - Improves consistency
- **Responsible radiology reporting:**
 - Report present and relevant absent findings
 - Describe at high granularity covering location, laterality, severity, etc.

Problems with Generative AI in Radiology

- Limited number of findings covered
- Diagnostic accuracy not sufficient for clinical use
 - **hallucinations** (fabricating or missing key findings), **laterality errors** (confusing left and right), and **diagnostic inaccuracies**
- Performance dominated by prevalent findings. Harder to catch rarer findings
- Opaque reasoning – not very explainable
- Much of the source of hallucination is due to auto-regressive language generation

Auto-read from AI has errors

Present findings: enlarged cardiac silhouette, pacer, calcifications, sternotomy wires, mediastinal clips, clear lungs

Absent findings: pleural effusion, vascular congestion

Ground truth report

MEDICAL CONDITION:

78 year old woman with chest pain, hx of chf

REASON FOR THIS EXAMINATION:

please evaluate for edema

FINAL REPORT

CHEST, TWO VIEWS:

HISTORY: 78-year-old female with chest pain. History of congestive failure. Question pulmonary edema.

FINDINGS: PA and lateral views of the chest are compared to previous exam from .

Compared with prior, there has been no significant interval change. The lungs remain clear. There is no pleural effusion. There is no pulmonary vascular engorgement. Cardiac silhouette is enlarged, but stable in configuration. Biventricular pacing device again seen with multiple leads stable positions. Atherosclerotic calcifications seen throughout the aorta. Median sternotomy wires and mediastinal clips again noted.

IMPRESSION: No acute cardiopulmonary process.



RRG models



Match to GT



Mismatches

X-ray GPT:

In this patient who has an **upper lobe nodule** and associated pain complaints, there is no indication of **pleural effusion** and **cardiac failure**. However, **edema** can be present, which should be evaluated to exclude a pulmonary disease. The patient has mild **pulmonary fibrosis** and may have **underlying COPD** which should also be considered. The findings suggest **cardiac failure** and the patient requires diuretics for treatment. The **ECG appears normal** and the dual lead AID **pacing device** remains intact.

RRRG:

The lungs are clear. No acute cardiopulmonary process. The **cardiomediastinal silhouette** and hilar contours are normal. There is **no focal consolidation, effusion, or pneumothorax**. There are **no acute osseous abnormalities**. **Moderate cardiomegaly** is unchanged. Left-sided **pacemaker leads** terminate in the right atrium and right ventricle.

GPT4o:

FINDINGS:

Lines/Tubes: **Pacemaker leads are in place.**

Lungs: **Mild interstitial markings observed; no acute infiltrates.**

Pleura: **No significant pleural effusion detected.**

Heart and Mediastinum: **Enlarged cardiac silhouette** consistent with known **CHF**.

Bones and soft tissues: **No acute fractures or bone abnormalities detected.**

Errors could be due to :

- Missed findings
- Incorrect or hallucinated findings
- Incorrect descriptions of findings
- Reversed findings (presence -> absence or vice versa)



IEEE
ISBI 2026

How to address such errors?

- Existing methods are mostly at training time
 - Alignment methods such as PPO, DPO
 - Used at training time
 - Human input
- Inference-time methods
 - Reinforcement learning methods: GRPO
 - Require generation of multiple outputs for reinforcement learning
 - Consensus could still present hallucinations
 - LLM as a judge
 - Still needs a ground truth
 - Suffers from hallucination as well
- Radiology needs:
 - Error detection and correction should be at inference time
 - Such detection and correction should work for any RRG model
 - No ground truth can be expected for correction
 - Inference-time operations should be lightweight for performance

Verification AI

- A class of designed to detect, measure, correct, and ultimately prevent errors in AI-generated clinical reports
- But how is this possible?
 - Since the output of generative AI could be a very large number of possibilities depending on the model, training prep, etc.
- This is possible in domain-specific contexts such as RRG
 - Reports describe known findings in the image.
 - Description syntax follows some common norms (e.g. presence or absence, core findings, location, laterality, etc.)
 - Errors made by RRG models are in the description of the findings
 - Could be generalizable for other domain-specific contests
- Such models could be discriminative or generative themselves

Verification AI models

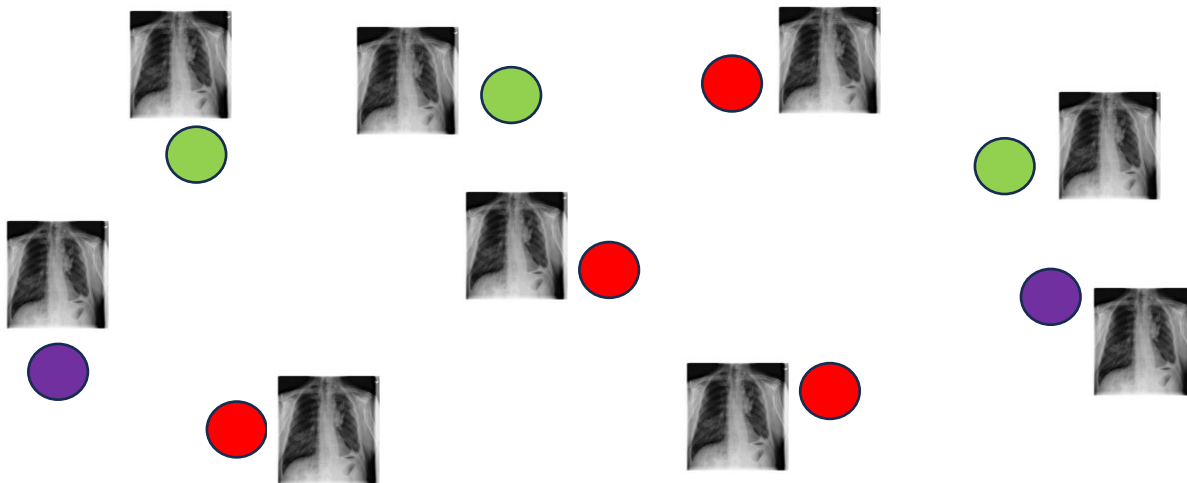
- Fact-checking discriminative models
 - Detects errors in the output produced by generative AI
- Report correction models
 - Corrects for the errors detected
- Non-RL Inference-time alignment models
 - Lightweight alignment of RRG models onsite using corrective models
- Retrieval-augmented verification
 - Trains LLM as judges through retrieval-augmented verification based on patient similarity

Modeling the space of errors made by generative AI

Collect labeled image-finding pairs simulating errors

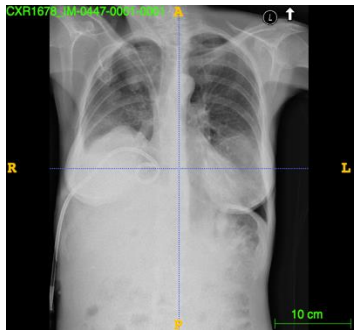
Synthesize Image-Text Data

- Representative of typical errors in automated reports
- Realistic and possible combinations verifiable through ground truth
- Robust to written variations



Synthesizing image-text pairs

Synthetic Data
Generation



- *Many many combinations possible!*
- *Not all of them are realistic.*
- *How to filter?*
 - *Based on findings provided*
 - *Avoid repetitions*
 - *Remove Contradictions*

R1

The heart is normal in size. There is no pneumothorax. There is again elevation of right hemidiaphragm with right-sided pleural effusion.



- Normal heart
- No pneumothorax
- Elevated hemidiaphragm
- Pleural effusion

Real

R2

The left lung appears grossly clear. Drainage catheter seen overlying the right upper quadrant.



- Clear lung
- Chest tube

Fake

R3

There are diffuse increased interstitial markings, suggestive of pulmonary fibrosis in bilateral lung. There is a small left pleural effusion/pneumothorax.



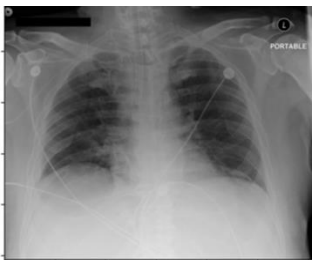
- ILD
- Pulmonary fibrosis
- Pleural effusion
- Pneumothorax

Drop pleural effusion
Drop pneumothorax

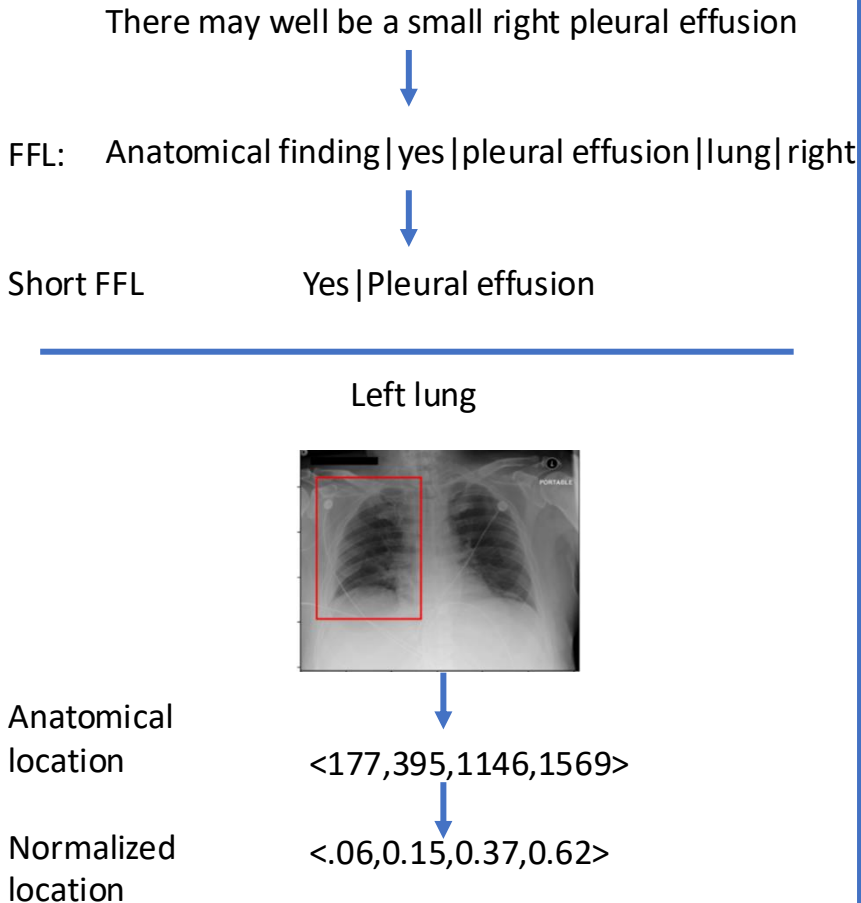


Synthesizing Data – Labeling findings

Synthetic Data
Generation



Mild basilar {atelectatic changes}
without evidence of acute
pneumonia or vascular
congestion. There may well be a
small right pleural effusion.



INPUT

Yes | Pleural
effusion

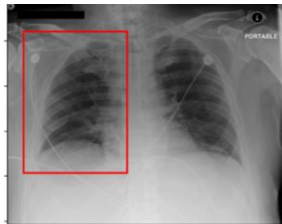
LABEL

$\langle .06, 0.15, 0.37, 0.62, 1 \rangle$

1=Real,
0=Fake

Synthesizing Data – Perturbing presence findings

Synthetic Data
Generation



Yes | Pleural effusion

$\langle .06, 0.15, 0.37, 0.62, 1 \rangle$

There may well be a small right
pleural effusion.

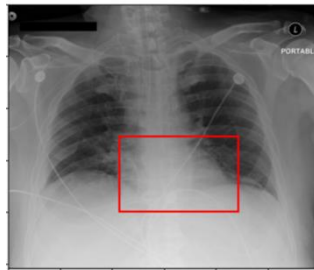
Finding is correct but location is incorrect



Yes | Pleural effusion

$\langle 0.47, 0.13, 0.38, 0.64, 0 \rangle$

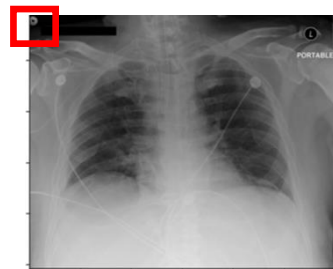
Exchange Finding



Yes | Cardiomegaly

$\langle 0.35, 0.5, 0.37, 0.28, 0 \rangle$

Reverse Finding

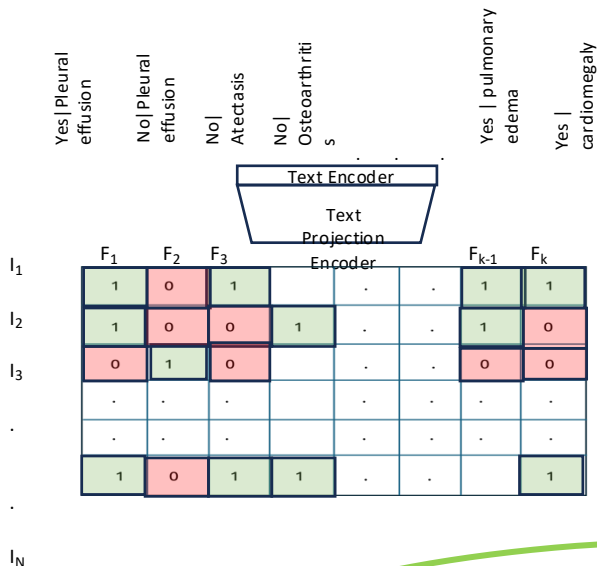
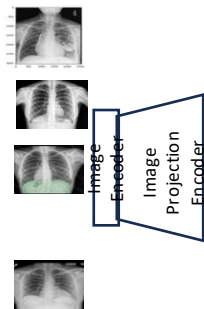


No | Pleural effusion

$\langle 0, 0, 0, 0, 0 \rangle$

Multi-label cross-modal supervised contrastive feature learning

Building
Fact-checking Model



Supervised contrastive CLIP (SC-CLIP)

- Sparse similarity matrix. Easy to train
- Pre-trained image and text encoders from ChexZero
- Projection encoders fresh-trained using new loss function
 - 768x512 for image
 - 512x512 for text
- 151,277,313 parameters

Cross-modal supervised
contrastive loss

$$\mathcal{L}_{SupC} = \sum_{i \in I} \frac{-1}{|F_{iReal}|} \sum_{f_{ij} \in F_{iReal}} \log \frac{e^{s_i f_{ij} / \tau}}{\sum_{a_{ik} \in F_{iFake}} e^{s_i a_{ik} / \tau}}$$

Real findings

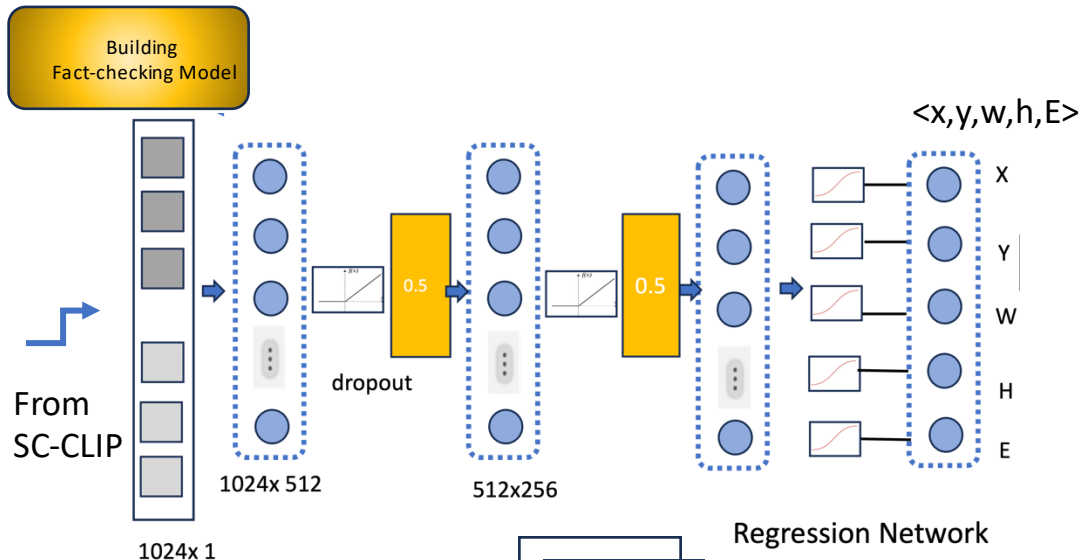
$s_i f_{ij} = z_i \cdot z_{f_{ij}}$ Latent features: real findings

$s_i a_{ik} = z_i \cdot z_{a_{ik}}$ Latent features: fake findings

Fake findings

Unlike SCL, the denominator is
over the fake findings per image

Cross-modal regression network



- Ground truth labels:
 $Y_g = \langle Y_{1g}, Y_{2g} \rangle$
 where $Y_{1g} = \langle x_g, y_g, w_g, h_g \rangle$ and $Y_{2g} = E_g = 1/0$
- Predicted labels:
 $Y_p = \langle Y_{1p}, Y_{2p} \rangle$
 where $Y_{1p} = \langle x_p, y_p, w_p, h_p \rangle$ and $Y_{2p} = E_p = 1/0$
- Convex hull of bounding boxes Y_{1g}, Y_{1p} $C_{Y_{1p}, Y_{1g}}$
- 657,413 parameters

$$\mathcal{L}_{Reg_i} = \underbrace{|Y_{1p} - Y_{1g}|}_{\mathcal{L}_1(Y_{1p}, Y_{1g})} + \underbrace{\frac{|Y_{1p} \cap Y_{1g}|}{|Y_{1p} \cup Y_{1g}|} - \frac{|C_{Y_{1p}, Y_{1g}} \setminus Y_{1p} \cap Y_{1g}|}{|C_{Y_{1p}, Y_{1g}}|}}_{\mathcal{L}_{giou}(Y_{1p}, Y_{1g})} + \underbrace{|Y_{1p} - Y_{1g}|^2}_{\mathcal{L}_{mse}(Y_{1p}, Y_{1g})} - \underbrace{[Y_{2g} \log(Y_{2p}) + (1 - Y_{2g}) \log(1 - Y_{2p})]}_{\mathcal{L}_{BCE}(Y_{2p}, Y_{2g})}$$

L1 loss for each spatial coordinate

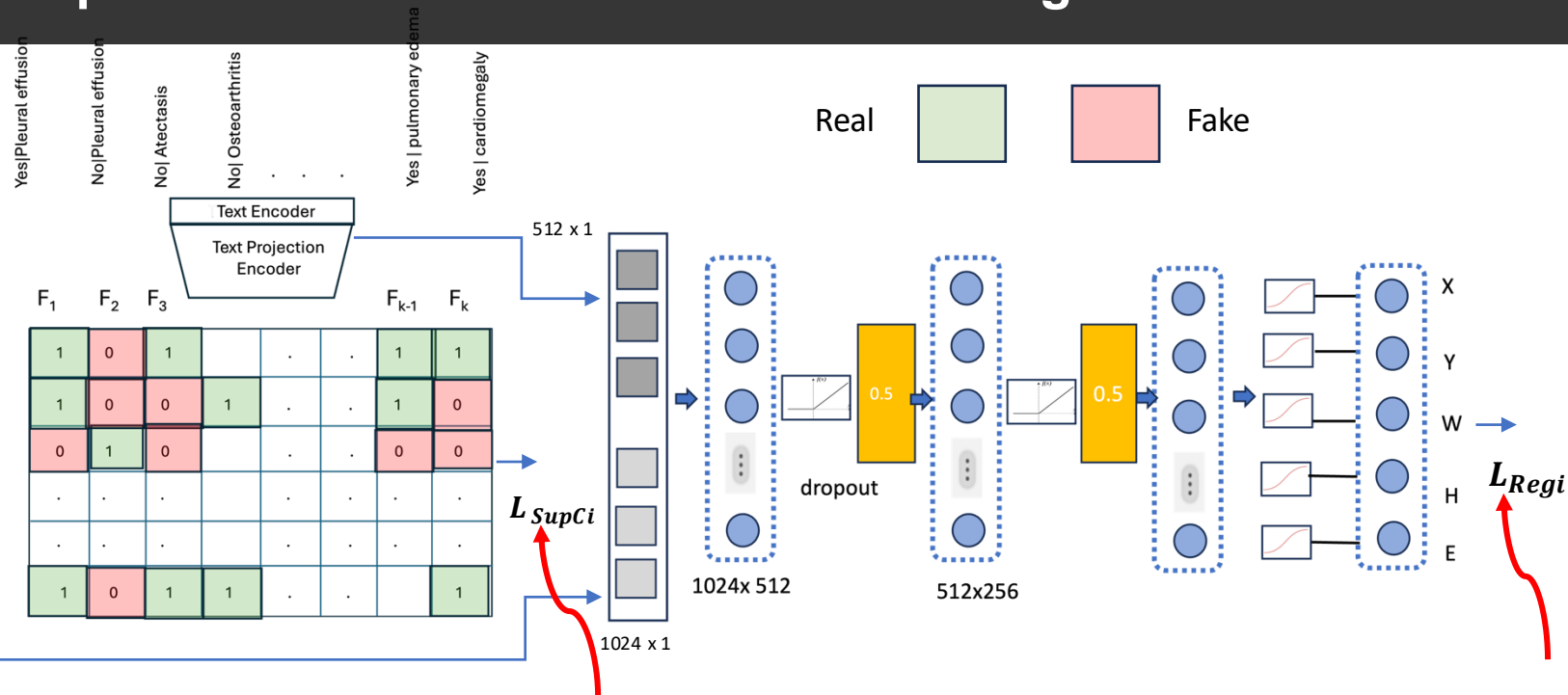
Bbox region overlap GIOU loss

MSE to account for absence findings $\langle 0, 0, 0, 0 \rangle$ coords

BCE loss for the finding veracity label

FCModel: Supervised Cross-modal Contrastive Regression Network

Building
Fact-checking Model



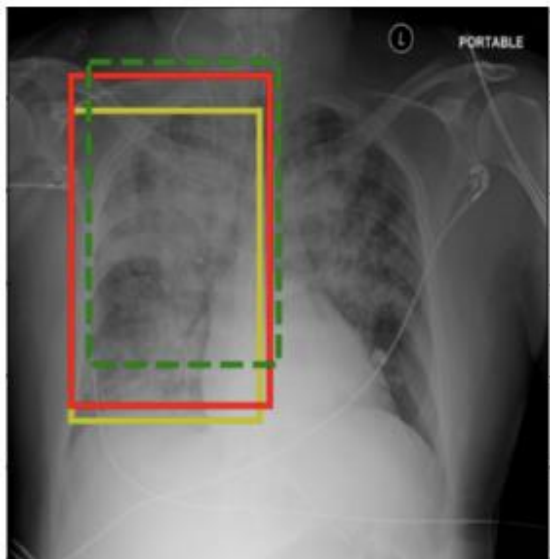
- Built as an end-to-end trainable network
 - image and text projection encoders
 - concatenated image-text representation
 - Regression on contrasted features
- Findings can be presence or absence findings

- Novel formulation differs from:
 - Contrastive learning
 - Supervised contrastive learning
 - Cross-modal CLIP-style learning
 - Multi-label contrastive learning

Explainable fact-checking Example – Correct case

Building
Fact-checking Model

Case: Finding and location are correct



Ground truth sentence

... pronounced alveolar opacities in a perihilar distribution, right greater than left, likely representing pulmonary edema

Automated report sentence

The pulmonary vasculature is not engorged, and the patient has moderate pulmonary edema on the right.

Target finding

Yes| Pulmonary edema

Implied merged anatomical locations

['right hilar structures', 'right lung']

GT location

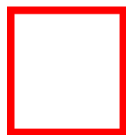
[0.120, 0.125, 0.366 0.586]

Predicted location

[0.151, 0.100, 0.351, 0.538, 0.428]

Predicted label

1



GT



Predicted



Indicated

Explainable fact-checking Example – Incorrect case

Building
Fact-checking Model

Case: Finding is correct but location is wrong

Ground truth sentence

Opacities are present at the lung bases bilaterally.

Automated report sentence

Subtle opacity in the right upper lung.

Target finding

Yes| Opacity

Implied merged anatomical locations

[right upper lung]

GT location

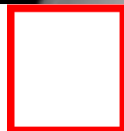
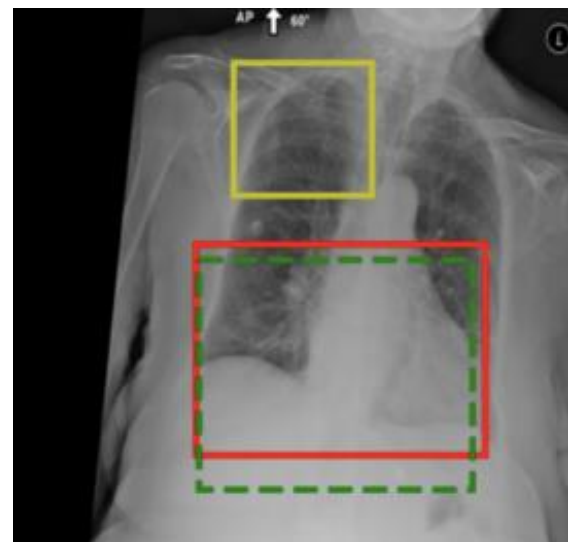
[0.328, 0.419, 0.498, 0.363]

Predicted location

[0.332, 0.442, 0.470, 0.396]

Predicted label

0



GT



Predicted



Indicated

Report correction performance

All 7 report generators evaluated against the test partitions of 4 datasets

Report Generator	CImaGenomeG		MS-CXR		ChestX-ray8		VinDR-CXR	
	<i>RQ</i>		<i>RQ</i>		<i>FC Score</i>		<i>RQ</i>	
	(A,P)	(A,G)	(A,P)	(A,G)	(A,P)	(A,G)	(A,P)	(A,G)
RGRG[65]	0.541	0.537	0.329	0.308	0.305	0.298	0.549	0.537
XrayGPT[66]	0.622	0.626	0.388	0.391	0.377	0.355	0.618	0.609
GPT4-inhouse	0.658	0.653	0.433	0.426	0.399	0.408	0.636	0.630
R2GenGPT[73]	0.587	0.585	0.377	0.374	0.346	0.333	0.581	0.579
CV2DistillGPT2[45]	0.576	0.573	0.439	0.433	0.427	0.420	0.588	0.6
CheXRepair[54]	0.744	0.733	0.466	0.461	0.439	0.432	0.709	0.714
Maira-2[4]	0.619	0.633	0.423	0.425	0.412	0.419	0.578	0.569

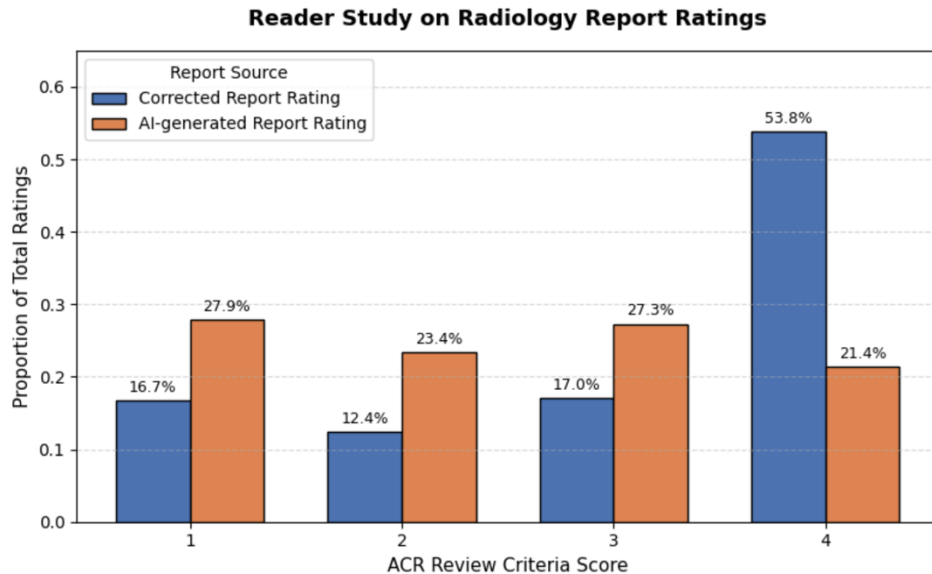
Generator	RadF1		RQ		BLEU		SBERT	
	(A,G)	(C,G)	(A,G)	(C,G)	(A,G)	(C,G)	(A,G)	(C,G)
RGRG[16]	0.52	0.67	0.46	0.52	0.24	0.29	0.33	0.43
XrayGPT[17]	0.39	0.45	0.37	0.48	0.14	0.24	0.26	0.38
GPT4-in	0.43	0.51	0.35	0.47	0.11	0.19	0.09	0.14
R2GenGPT[18]	0.54	0.58	0.37	0.49	0.19	0.27	0.38	0.47
CV2GPT2[19]	0.41	0.49	0.38	0.48	0.14	0.24	0.43	0.54
CheXRepair[3]	0.38	0.43	0.36	0.44	0.21	0.28	0.39	0.46
Maira-2[2]	0.58	0.63	0.52	0.59	0.20	0.26	0.43	0.51
Avg.Improv.	13.5%		27%		48.2%		32.5%	

Model	CCC (A,P)(A,G)
FCModel	0.997
FCSimple	0.831

FC model has high CCC!

MIMC Reader Study

- Compare qualitative difference:
 - corrected and generative AI report
- Test dataset:
 - 160 images from CG-Gold
- Report generators evaluated:
 - XrayGPT
 - GPT4-o
 - ChexAgent
- 3 board-certified clinicians
- Blinded to report source
- ACR Review criteria

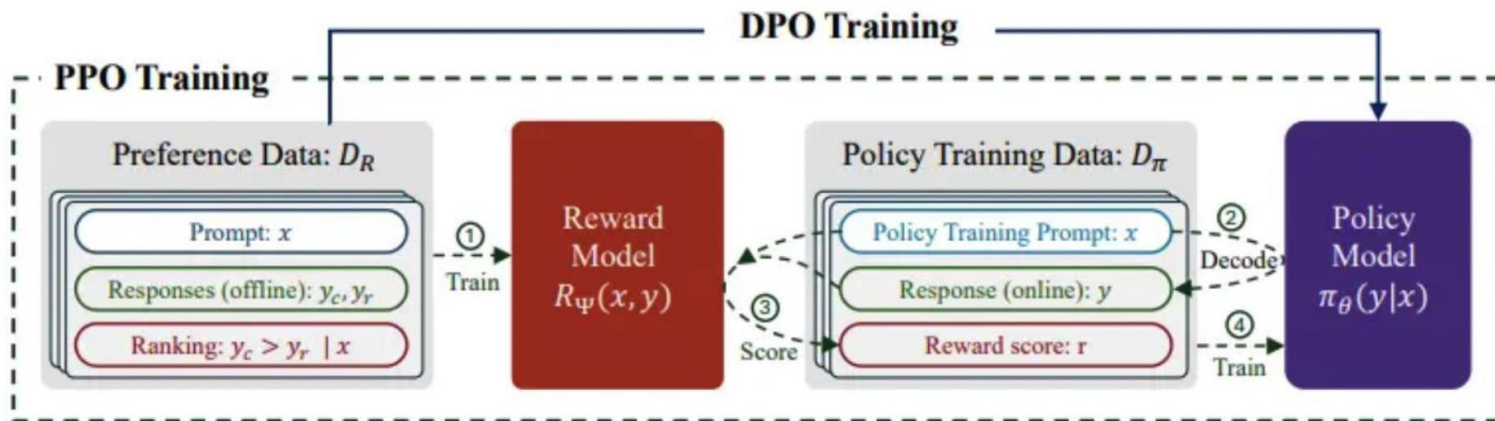


Using verification models for inference-time alignment

- Improve VLM models through alignment post-training
 - PPO, DPO, GRPO
 - Already being done using manual labels or ground truth reports
- But if these models are already deployed?
 - Can lightweight onsite fine-tuning help?
 - Yes:
 - But no ground truth data is available to correct the output
 - GRPO then relies on consensus leading to false negatives
 - LLM-as-judge are also a problem since none is a 'correct' VLM

Direct policy optimization (DPO)

- No separate reward model needed
- Implicit reward included in the policy update rule.
 - Analytical mapping derived from reward functions to optimal policies, which transform a loss function over reward functions into a loss function over policies
- Human feedback still taken for preferences after SFT of the LLM
- The LLM model is optimized without using RL by minimizing the L_{DPO} loss.



[Direct Preference Optimization: Your Language Model is Secretly a Reward Model, NeurIPS'23](#)

Inference-time alignment

- Based on findings
 - They are key to determining if a report is correct
 - Also used in EditGRPO approach
 - Our approach extracts both names and geometric locations of findings
- Using a fact-checking model supplies surrogate ground truth
- Forms a corrected report for updating the policy
 - The policy update rule requires textual response as the preferred response.
- Similar approach to direct preference optimization (DPO) done at training time

Direct preference optimization

- Not an RL method
 - a frozen reference policy π_{ref}
 - a trainable policy π_{θ}

Human-curated preference
training data

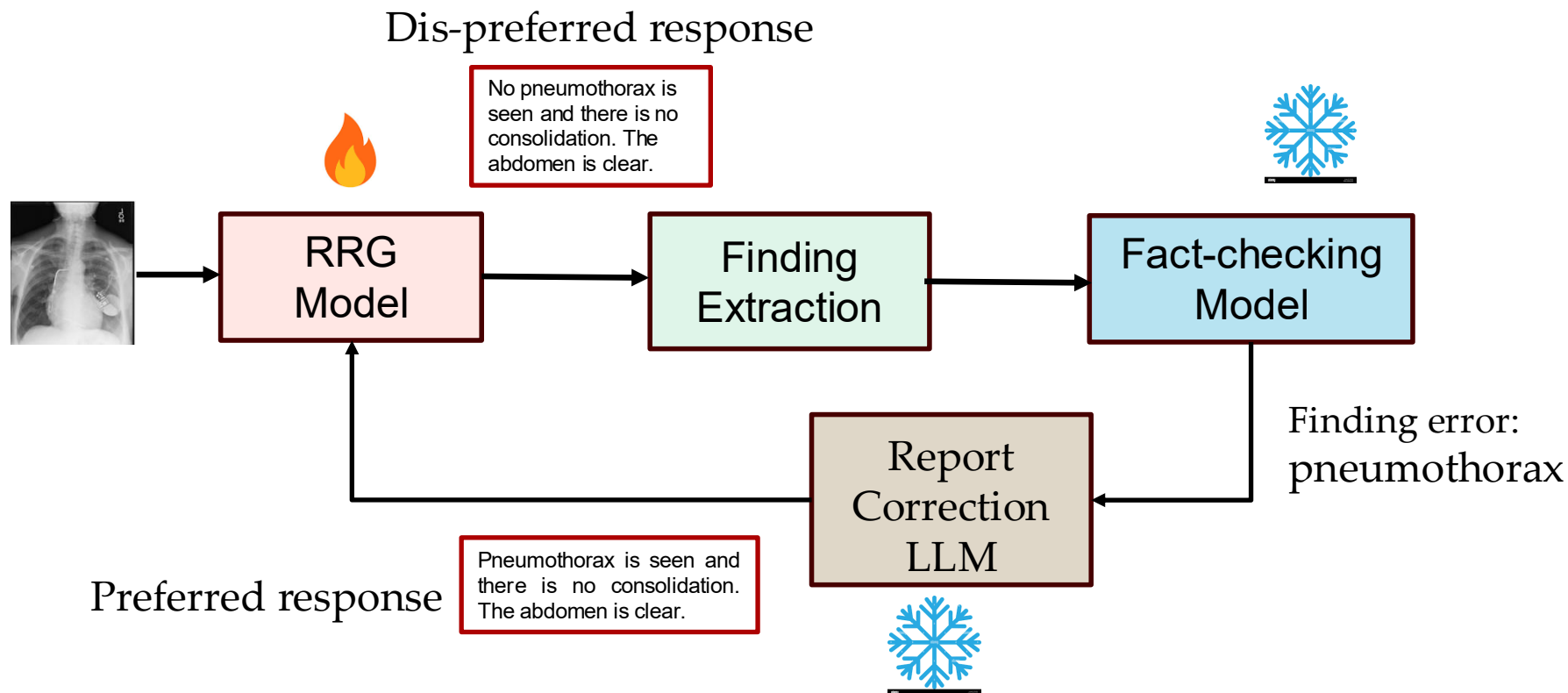
(x, y^+, y^-)

Maximize

$$\log \sigma \left(\beta \left[\log \pi_{\theta}(y^+|x) - \log \pi_{\theta}(y^-|x) - (\log \pi_{\text{ref}}(y^+|x) - \log \pi_{\text{ref}}(y^-|x)) \right] \right)$$

- Increase probability of preferred answer
- Decrease probability of rejected answer
- Relative to the reference policy
- offline, supervised optimization on preference pairs.
- Still needs preference pairs to be assembled

Automatic preference optimization (APO) – A new inference-time alignment



Preference alignment method

Create a phrase-grounded preference dataset from RRG model output

$$D = \{x^i, y_w^i, L_w^i, y_l^i, L_l^i\}_{i=1}^N$$

$$L_{APO}(\pi_\theta, \pi_{ref}) = -\mathbb{E}(x, y_w, y_l) \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)} + \gamma \log \frac{O_\theta}{O_{ref}} \right) \right]$$



KL-Divergence of
preferred response
with reference



KL-Divergence of
dispreferred response
with reference



Phrasal
grounding loss

π_{ref} : Original policy

π_θ : Updated policy

$x^{\{i\}}$ = initial prompt + Image

$y_w^{\{i\}}$ = preferred response

$y_l^{\{i\}}$ = dis-preferred response

M_w = Preferred response findings

M_l = Dispreferred response findings

M_{ref} = Original response findings

$L_l = \{l_j\}_{j=1}^{M_l} = \{ \langle u_j, v_j, w_j, h_j \rangle \}_{j=1}^{M_l}$

$L_w = \{l_j\}_{j=1}^{M_w} = \{ \langle u_j, v_j, w_j, h_j \rangle \}_{j=1}^{M_w}$

$$O_{ref} = \frac{1}{M_{ref}} \sum_j \frac{|l_{wrefj} \cap l_{lrefj}|}{|l_{wrefj} \cup l_{lrefj}|}$$

$$O_\theta = \frac{1}{M_\theta} \sum_j \frac{|l_{w\theta j} \cap l_{l\theta j}|}{|l_{w\theta j} \cup l_{l\theta j}|}$$

APO updates policy:

- Without deviating too much from the original response
- Keeping the corrected findings
- Increasing the overlap between the locations of corresponding findings

Results

- **17-40% improvement in quality**
- Multiple RRG models
- Multiple error metrics to judge quality improvement
- Comparison with DPO, PPO, GRPO
- Comprehensive coverage of findings

Generator	RadgraphF1			RQ			BLEU			SBERT			Green		
	(A,G)	(P,G)	(C,G)	(A,G)	(P,G)	(C,G)	(A,G)	(P,G)	(C,G)	(A,G)	(P,G)	(C,G)	(A,G)	(P,G)	(C,G)
RGRG[33]	0.52	0.67	0.69	0.46	0.52	0.56	0.24	0.29	0.34	0.33	0.43	0.46	0.39	0.43	0.45
XrayGPT[34]	0.39	0.45	0.47	0.37	0.48	0.52	0.14	0.24	0.28	0.26	0.38	0.43	0.28	0.32	0.36
GPT4-in	0.43	0.51	0.54	0.35	0.47	0.51	0.11	0.19	0.24	0.09	0.14	0.19	0.33	0.36	0.45
R2GenGPT[36]	0.54	0.58	0.59	0.37	0.49	0.53	0.19	0.27	0.34	0.38	0.47	0.50	0.29	0.34	0.37
CV2GPT2[20]	0.41	0.49	0.51	0.38	0.48	0.54	0.14	0.24	0.29	0.43	0.54	0.56	0.30	0.34	0.37
CheXRepair[27]	0.38	0.43	0.44	0.36	0.44	0.49	0.21	0.28	0.34	0.39	0.46	0.49	0.32	0.34	0.39
Maira-2[1]	0.58	0.63	0.66	0.52	0.59	0.61	0.20	0.26	0.29	0.43	0.51	0.54	0.43	0.47	0.49
Avg.Improv.	13.5%			20.2%			27%			35.0%			48.2%		

Generator	MCXR			C8			Vindr		
	RadF1	SBERT	Green	RadF1	SBERT	Green	RadF1	SBERT	Green
RGRG	0.46/0.49	0.38/0.45	0.40/0.47	0.32/0.43	0.36/0.45	0.36/0.42	0.41/0.52	0.39/0.45	0.41/0.48
XrayGPT	0.37/0.40	0.32/0.41	0.34/0.39	0.38/0.43	0.31/0.40	0.28/0.36	0.32/0.40	0.38/0.46	0.39/0.45
GPT4-in	0.43/0.48	0.39/0.44	0.41/0.46	0.33/0.42	0.37/0.45	0.35/0.43	0.42/0.54	0.40/0.47	0.40/0.49
R2GenGPT	0.49/0.55	0.40/0.47	0.41/0.48	0.39/0.43	0.40/0.47	0.42/0.50	0.40/0.48	0.42/0.47	0.42/0.49
CV2GPT2	0.40/0.44	0.37/0.43	0.39/0.43	0.33/0.40	0.34/0.40	0.35/0.45	0.42/0.48	0.39/0.46	0.41/0.49
CheXRepair	0.37/0.40	0.34/0.41	0.38/0.41	0.32/0.39	0.32/0.41	0.32/0.40	0.38/0.45	0.31/0.44	0.41/0.48
Maira-2	0.49/0.56	0.43/0.59	0.51/0.57	0.39/0.47	0.49/0.56	0.51/0.58	0.39/0.48	0.45/0.56	0.42/0.54
Avg. Impv	8.9%	16.8%	13.1%	21.2%	10.7%	10.1%	13.3%	11.1%	9.5%

Generator	CG				MCXR				C8				VinDr				ChexPert			
	RQ				RQ				RQ				RQ				RQ			
	(A,G)	(C,G)	(D,G)	(R,G)	(A,G)	(C,G)	(D,G)	(R,G)	(A,G)	(C,G)	(D,G)	(R,G)	(A,G)	(C,G)	(D,G)	(R,G)	(A,G)	(C,G)	(D,G)	(R,G)
RGRG[33]	0.46	0.56	0.51	0.46	0.51	0.62	0.57	0.52	0.38	0.49	0.43	0.39	0.51	0.63	0.57	0.51	0.43	0.54	0.50	0.44
XrayGPT[34]	0.37	0.52	0.46	0.36	0.45	0.49	0.46	0.44	0.35	0.42	0.38	0.40	0.46	0.54	0.49	0.45	0.49	0.54	0.51	0.48
GPT4-inhouse	0.35	0.51	0.46	0.38	0.46	0.54	0.51	0.46	0.41	0.48	0.43	0.42	0.51	0.58	0.54	0.51	0.47	0.54	0.51	0.47
R2GenGPT[36]	0.37	0.53	0.43	0.38	0.44	0.54	0.48	0.45	0.38	0.47	0.42	0.39	0.51	0.57	0.54	0.52	0.51	0.61	0.55	0.52
CV2DisGPT2[20]	0.38	0.54	0.45	0.38	0.39	0.49	0.43	0.38	0.41	0.47	0.44	0.42	0.52	0.6	0.56	0.53	0.48	0.59	0.55	0.47
CheXRepair[27]	0.36	0.49	0.43	0.35	0.45	0.51	0.48	0.45	0.43	0.49	0.46	0.44	0.51	0.59	0.56	0.50	0.49	0.59	0.52	0.49
Maira-2[1]	0.52	0.61	0.57	0.51	0.47	0.58	0.53	0.48	0.41	0.49	0.45	0.42	0.50	0.61	0.57	0.51	0.52	0.61	0.56	0.53
Avg. Impv.	35.2%				18.6%				0.5%				18.9%				9.1%			

Summary

- Verification AI is about discriminative and generative models focused on detecting, correcting and fine-tuning generative AI models at inference time.
- Current generative AI offers no guarantees of performance
- Using Verification AI methods in combination with generative AI could lead towards more trustworthy AI solutions in future.
- RadCheck – A synthetic data set of 27 million pairs of image and finding in HuggingFace

Where is the field headed?

- Foundational models still have limitations
 - Not truly domain independent despite the large knowledge given
 - Reasoning powers are still limited.
 - Largely unsuccessful in healthcare world
- New thinking is needed for AI to make a larger impact.
 - Domain understanding
 - New datasets interdisciplinary emerging
 - Domain reasoning
 - New architectures
 - From just scaling models to architectural efficiency, better compute/memory, and flexible designs for innovation
- Move from larger to smaller models
- Ethical concerns still looming
- Real cognition still far away